

Moving nursing beyond $p < .05$



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Presenters

Matthew Hayat, PhD

Professor of Biostatistics
School of Public Health
Byrdine F. Lewis College of Nursing and Health Professions
Georgia State University



Martina Mueller, PhD

Professor of Biostatistics
College of Nursing
Department of Public Health Sciences (Joint)
Medical University of South Carolina



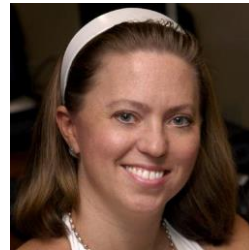
Mary Dooley, MS

Research Associate
College of Nursing
Medical University
of South Carolina



Melinda Higgins, PhD

Research Professor of Biostatistics
School of Nursing
Emory University



Michael Weaver, RN, PhD, FAAN

Professor of Biostatistics
Associate Dean for Research and Scholarship
College of Nursing
University of Florida



Outline

- I. Statistics in Nursing Research and Education
- II. Classical Hypothesis & Significance Testing
- III. The ASA Movement
- IV. Recommendations for Change

Statistics in Nursing Research and Education

Background

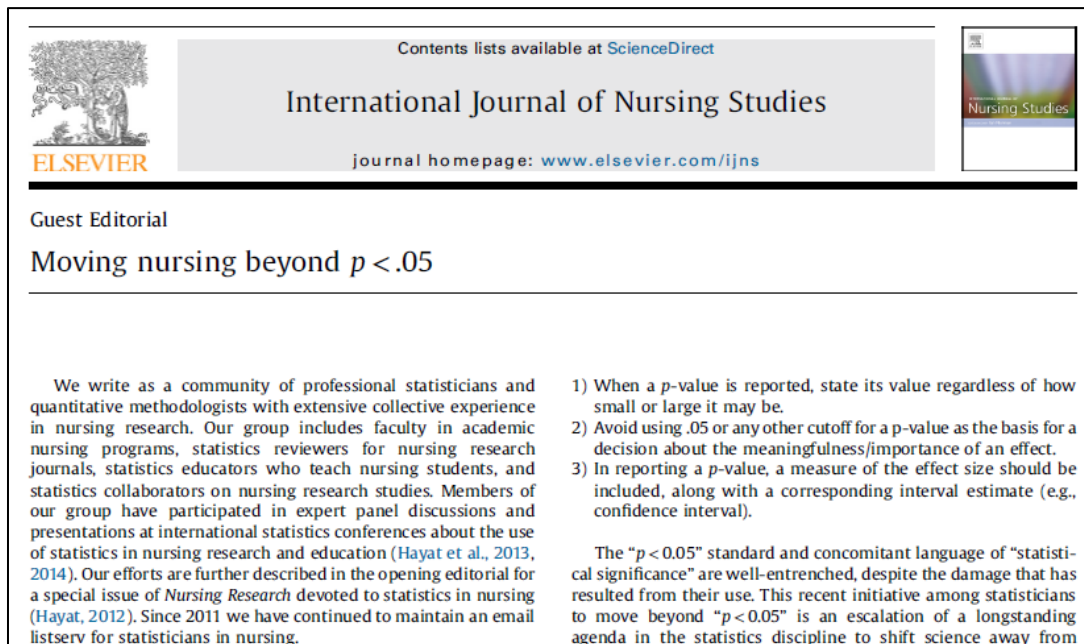
- We are members of a community of statisticians and quantitative methodologists involved in nursing education and research
- Our group includes faculty in academic nursing programs, statistics reviewers for nursing research journals, statistics educators who teach nursing students, and statistics collaborators on nursing research studies
- Members of our group have participated in expert panel discussions and presentations at international statistics conferences about the use of statistics in nursing research and education

-Add about diversity of members...

Statistics in Nursing Education and Research

Moving Nursing Beyond $p < .05$

Our work was recently published as a guest editorial:



This has also been published in ***Nursing Outlook, Research in Nursing & Health, Research and Theory for Nursing Practice***

Hayat, M.J., Staggs, V.S., Schwartz, T.A., Higgins, M., Azuero, A., Budhathoki, C., Chandrasekhar, R., Cook, P., Cramer, E., Dietrich, M.S., Garnier-Villarreal, M., Hanlon, A., He, J., Hu, J., Kim, M., Mueller, M., Nolan, J.R., Perkhounkova, Y., Rothers, J., Schluck, G., Su, X., Templin, T.N., Weaver, M.T., Yang, Q., Ye, S. (2019). Moving nursing beyond $p < .05$. *International Journal of Nursing Studies*.

The ASA Movement



How did we get here?



Problems with p-values



Guidance and advice from the ASA



What to do next

1925: Fisher, R.A. (1925), Statistical Methods for Research Workers, Edinburgh: Oliver & Boyd.

DISTRIBUTIONS

45

only once in 370 trials, while Table II. shows that to exceed the standard deviation sixfold would need nearly a thousand million trials. The value for which $P = .05$, or 1 in 20, is 1.96 or nearly 2 ; it is convenient to take this point as a limit in judging whether a deviation is to be considered significant or not. Deviations exceeding twice the standard deviation are thus formally regarded as significant. Using this criterion, we should be led to follow up a false indication only once in 22 trials, even if the statistics were the only guide available. Small effects will still escape notice if the data are insufficiently numerous to bring them out, but no lowering of the standard of significance would meet this difficulty.

And so it began ...

So, why do/did we do it?

In February 2014, George Cobb, Professor Emeritus of Mathematics and Statistics at Mount Holyoke College, posed these questions to an ASA discussion forum:

Q: Why do so many colleges and grad schools teach $p = 0.05$?

A: Because that's still what the scientific community and journal editors use.

Q: Why do so many people still use $p = 0.05$?

A: Because that's what they were taught in college or grad school.



... and on and on
the vicious spiral
continued ...

Warnings about p-values have been around a long time (100+ years)

Regardless of whether it was ever useful, a declaration of “statistical significance” has today become meaningless. Made broadly known by Fisher’s use of the phrase (1925), Edgeworth’s (1885) original intention for statistical significance was simply as a tool to indicate when a result warrants further scrutiny. But that idea has been irretrievably lost. Statistical significance was never meant to imply scientific importance, and the confusion of the two was decried soon after its widespread use (Boring 1919). Yet a full century later the confusion persists.

What exactly is a p-value?

- Probability that the computed statistic based on the observed data is larger than what would be expected given that the NULL hypothesis is true

$P(\text{statistic from data} > \text{critical value} \mid H_0)$

- This tell us **NOTHING** about the data quality, effect size, importance, random chance, whether the null (H_0) hypothesis is true, ...

The trouble with p-values



From “The trouble with tribbles” Star-trek episode 15, season 2, 1967

<u>P-VALUE</u>	<u>INTERPRETATION</u>
0.001	HIGHLY SIGNIFICANT
0.01	
0.02	
0.03	
0.04	SIGNIFICANT
0.049	
0.050	OH CRAP. REDO CALCULATIONS.
0.051	ON THE EDGE OF SIGNIFICANCE
0.06	
0.07	HIGHLY SUGGESTIVE, SIGNIFICANT AT THE $P < 0.10$ LEVEL
0.08	
0.09	
0.099	HEY, LOOK AT THIS INTERESTING SUBGROUP ANALYSIS
≥ 0.1	

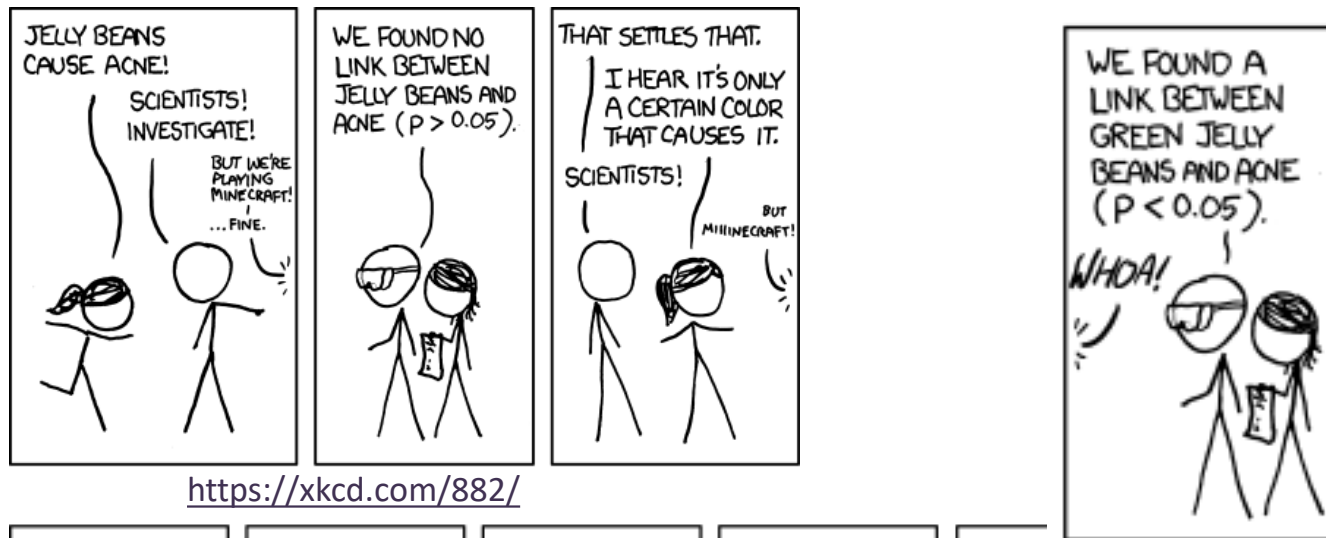
<https://xkcd.com/1478/>

YEA! Published!!

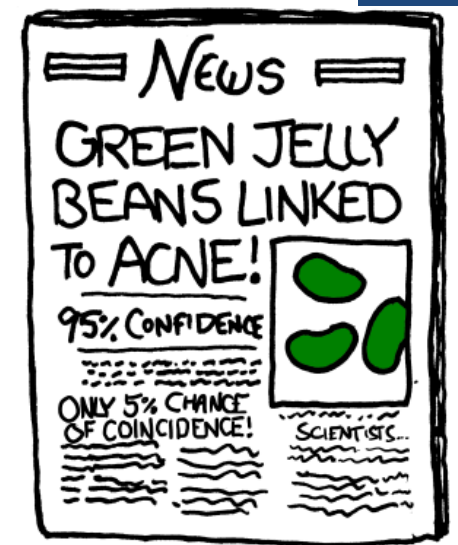
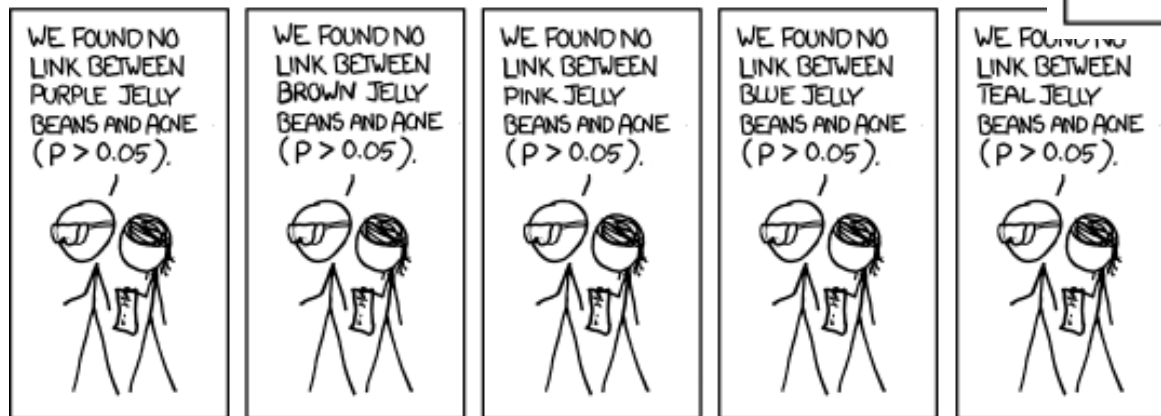
Grey Area...

BUMMER! Not Published??

And publication bias was born...

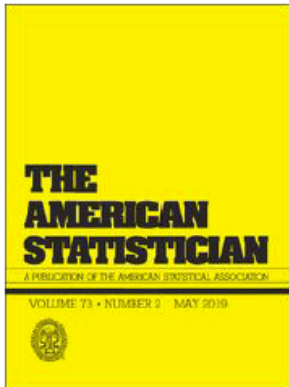


<https://xkcd.com/882/>



19 colors found $p > .05$, not significant, except for ...

P-hacking



The American Statistician

2016

ISSN: 0003-1305 (Print) 1537-2731 (Online) Journal homepage: <https://amstat.tandfonline.com/loi/utas20>

The ASA Statement on p -Values: Context, Process, and Purpose

Ronald L. Wasserstein & Nicole A. Lazar

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To link to this article: <https://doi.org/10.1080/00031305.2016.1154108>

ASA 6 Principles for p-values



DO indicate data incompatible with model



DO NOT measure probability hypothesis is true



Should NOT be used as basis for policy decisions

ASA 6 Principles for p-values



Require full reporting and transparency (for proper inference)



Is NOT an effect size, nor indication of importance



Is NOT a measure of evidence

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ISSN: 0003-1305 (Print) 1537-2731 (Online) Journal homepage: <https://www.tandfonline.com/loi/utas20>

Moving to a World Beyond " $p < 0.05$ "

Ronald L. Wasserstein, Allen L. Schirm & Nicole A. Lazar

To cite this article: Ronald L. Wasserstein, Allen L. Schirm & Nicole A. Lazar (2019) Moving to a World Beyond " $p < 0.05$ ", The American Statistician, 73:sup1, 1-19, DOI: [10.1080/00031305.2019.1583913](https://doi.org/10.1080/00031305.2019.1583913)

To link to this article: <https://doi.org/10.1080/00031305.2019.1583913>

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Statistical Inference in the 21st Century: A World Beyond $p < 0.05$

<https://www.tandfonline.com/toc/utas20/73/sup1?nav=tocList>

Open Access – many written with practical advice but some with more in depth technical details

ASA - Don't's

- Don't base conclusions solely on “statistically significant”
- Don't believe that an association exists just because it was statistically significant.
- Don't believe that an association is absent just because it was not statistically significant.

ASA - Don't's

- Don't believe that the p-value is the probability that the hypothesis is true.
- Don't conclude scientific importance based on statistical significance
- **DON'T SAY "STATISTICALLY SIGNIFICANT"**

STOP !

STOP
Saying/Writing

“Statistically
Significant”

2. Don't Say “Statistically Significant”

The *ASA Statement on P-Values and Statistical Significance* stopped just short of recommending that declarations of “statistical significance” be abandoned. We take that step here. We conclude, based on our review of the articles in this special issue and the broader literature, that it is time to stop using the term “statistically significant” entirely. Nor should variants such as “significantly different,” “ $p < 0.05$,” and “nonsignificant” survive, whether expressed in words, by asterisks in a table, or in some other way.

However, still report p-values, BUT don't use dichotomous decision making ...

There is
hope



Don't jump off a cliff just yet...

P-values are a result not the end goal



Common sense

- meaningful differences
 - A1c reduced from 12 to 8.4
 - We see an improvement of 30%
- meaningful thresholds
 - A1c >> 6 – still diabetes

It's not all about $p < .05$

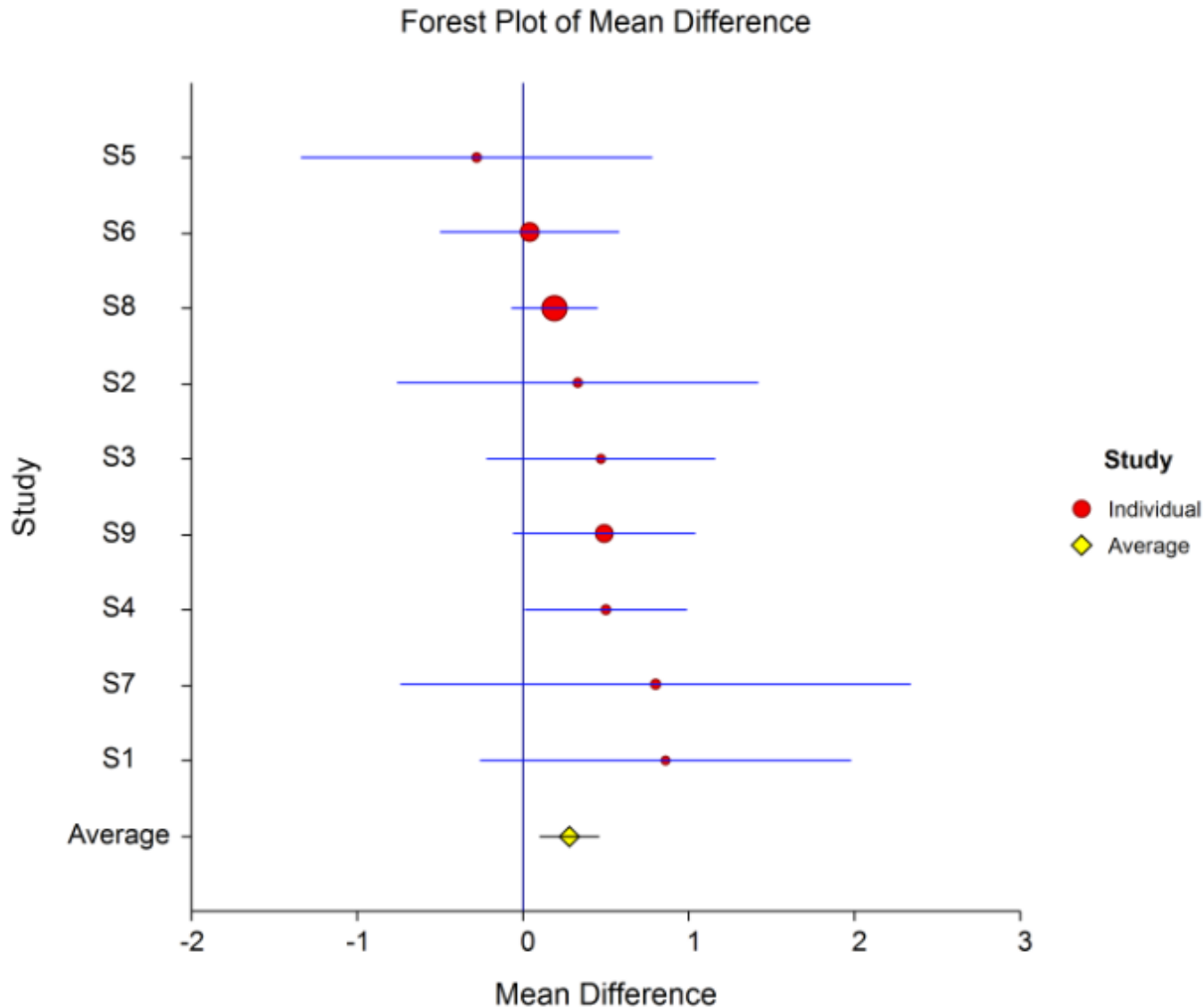
- sample size – would you change practice
 - based on 15 or 20 observations?
 - based on only one or a few studies?



Case(s) in Point

- Question: Is subcutaneous heparin more effective than intravenous heparin (initial treatment of DVT)?
- Hommes, et.al. (1992) *Annals of Internal Medicine*, 116: 279-284
 - Odds Ratio 0.62
 - 95% CI [0.39, 0.98]
 - $p < 0.05$: Yes, subcutaneous heparin more effective than intravenous heparin
- Messori, et.al. (1993) *Annals of Internal Medicine*, 118: 77-78.
 - Odds Ratio 0.61
 - 95% CI [0.298, 1.251]
 - $p > 0.05$, No, subcutaneous heparin is not (significantly) more effective than intravenous heparin
- Findings are “opposite” (in statistical significance, but OR similar)

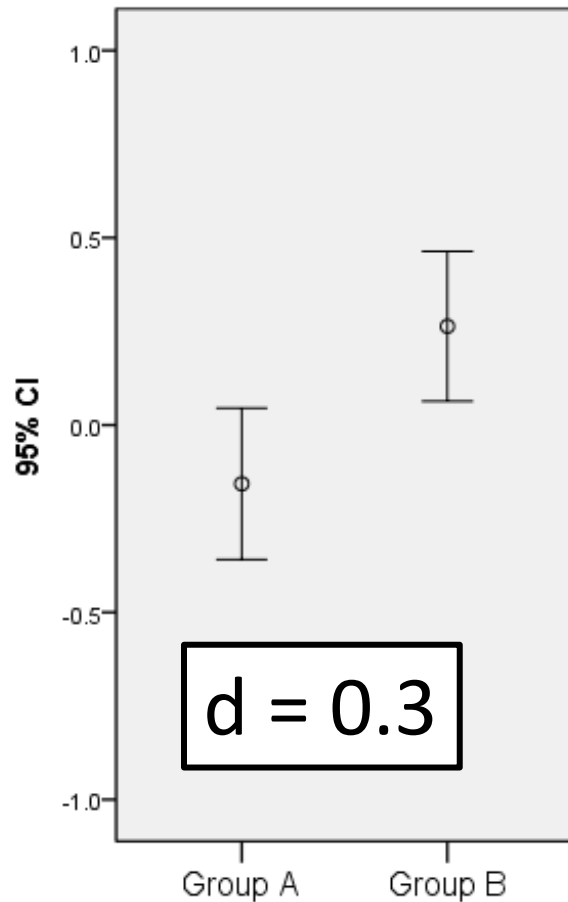
Meta-Analysis – multiple studies



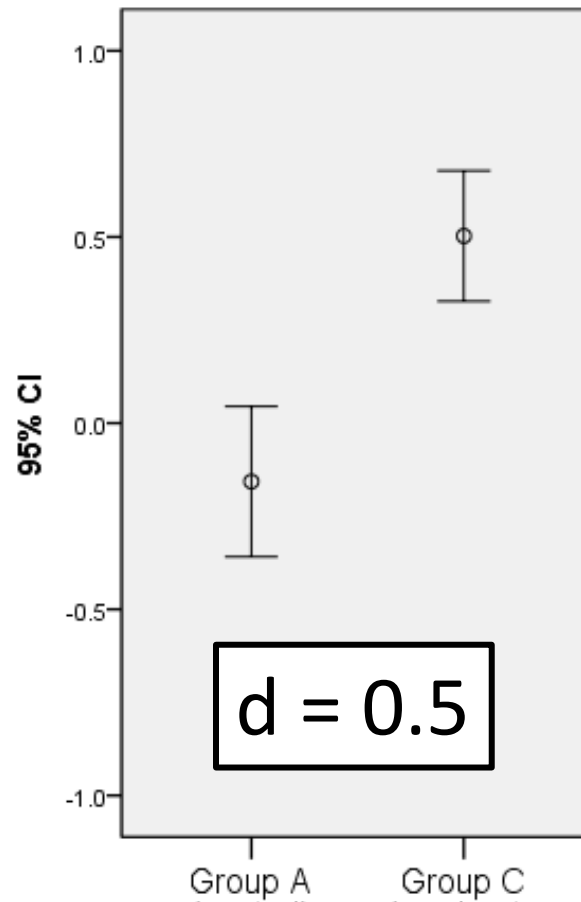
It's not all about $p < .05$: Effect Size

How big is the difference relative to the variability?

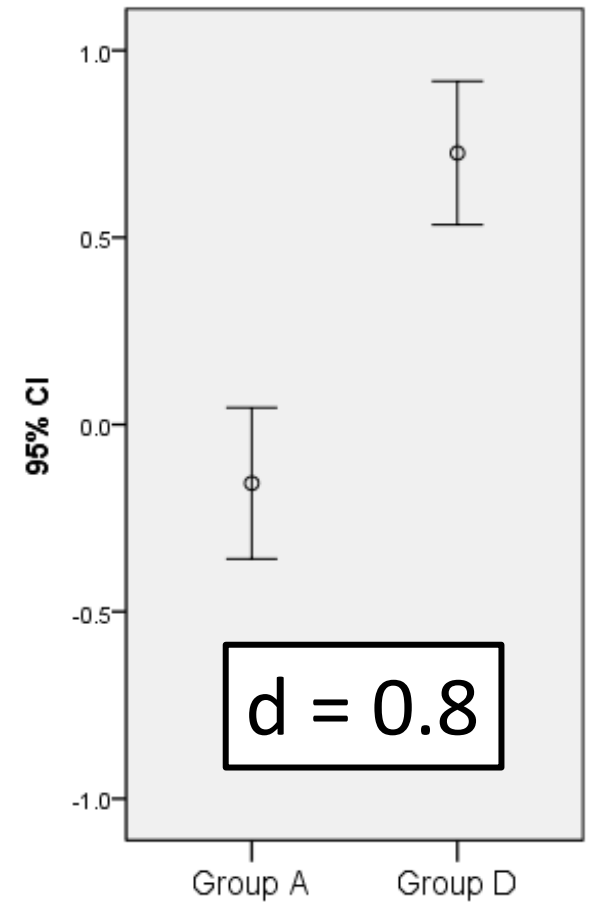
SMALL



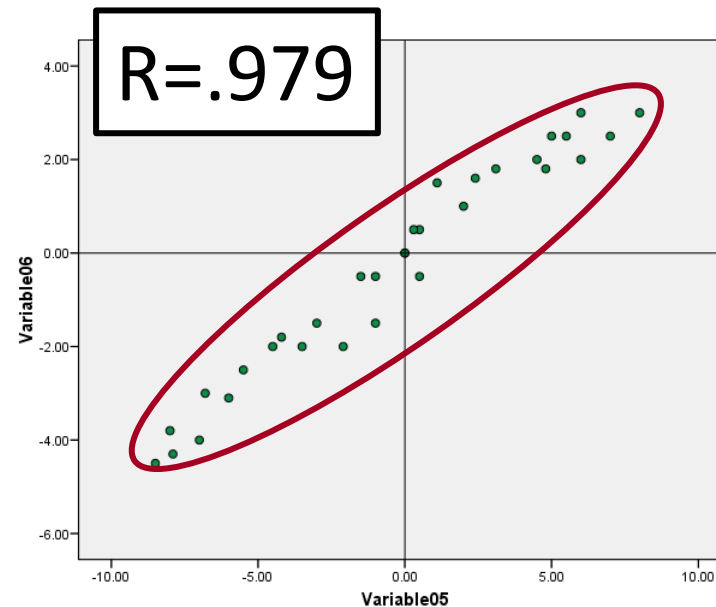
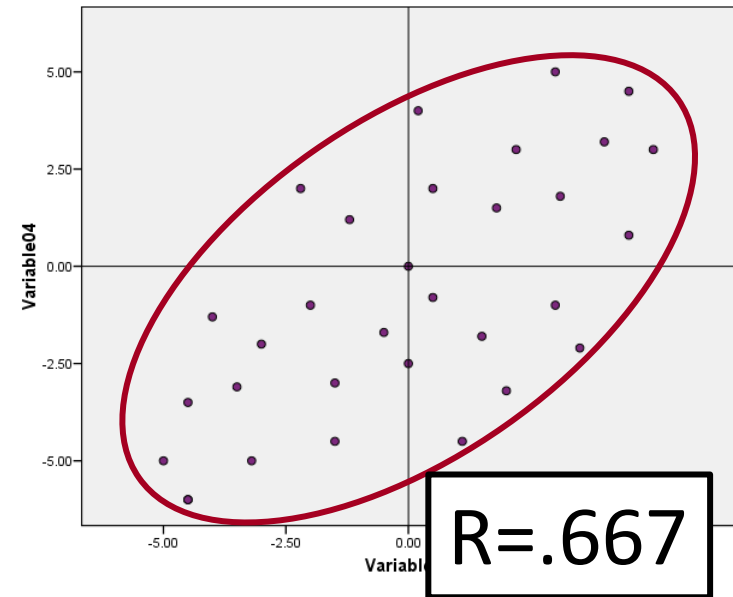
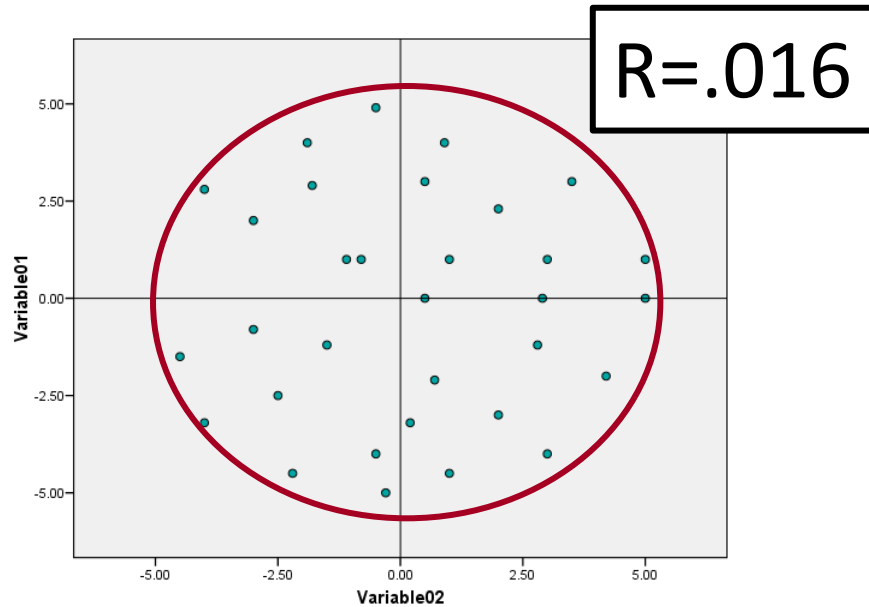
MEDIUM



LARGE



Effect Size: Correlation



Effect Sizes - Examples

Statistical Test or Model	Effect Size	Small	Medium	Large
T-test	Cohen's d	0.2	0.5	0.8
Correlation	r	0.1	0.3	0.5
Chi-square test	ω	0.1	0.3	0.5
ANOVA	Cohen's f	0.1	0.25	0.4
Linear Regression Model	r ²	0.01	0.09	0.25
Linear Regression Coefficients	Standardized Beta β	0.1	0.3	0.5
Logistic Regression	Odds Ratio	1.4	3.6	4.3

What to do?



EFFECT SIZES



**CONFIDENCE
INTERVALS**



**PREDICTION
INTERVALS**



**INTERPRET
AGAINST
THRESHOLDS**



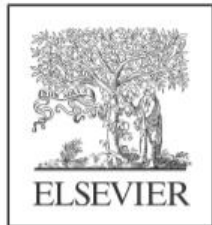
**CLINICALLY
MEANINGFUL**



**SCIENTIFIC
IMPORTANCE**

Recent Example – Nursing Outlook

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NURS OUTLOOK 000 (2019) 1–13

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The Community Resiliency Model[®] to promote nurse well-being

Linda Grabbe, PhD, FNP-BC, PMHNP-BC^{a,*}, Melinda K. Higgins, PhD^a,
Marianne Baird, MN, RN, ACNS-BC^b, Patricia Ann Craven, BSN, RN^c,
Sarah San Fratello, BSN, RN, CCRN^b

^aNell Hodgson Woodruff School of Nursing, Emory University, Atlanta, GA

^bEmory Healthcare, Atlanta, GA

^cChildren's Healthcare of Atlanta, Atlanta, GA

Given recent communications from the American Statistics Association (ASA), *p*-values for statistical tests and models are reported, however, emphasis has been placed on reporting and interpreting effect sizes and clinically descriptive differences (Wasserstein, Schirm, & Lazar, 2019). So, in addition to performing statistical models and tests with reported *p*-values, effect sizes (Cohen's *d*) were also computed based on the change scores from baseline to each follow-up time point (Cohen, 1988) to evaluate small (*d* = 0.2), moderate (*d* = 0.5), and large (*d* = 0.8) effect sizes to help determine clinically meaningful improvements. The percentages of subjects whose scores improved from baseline were also reported. All computations

Table 2 – Outcomes Over Time by Group

Outcome	Time	N	Mean (SD)	Change from Baseline ¹ Mean (SD)	Effect Size Cohen's <i>d</i>	Improved Percent (%)
Well-being CRM	Baseline	98	57.06 (18.62)			
	1 week	33	63.76 (13.45)	7.18 (10.88)	0.66	63.6%
	3 months	32	63.72 (18.82)	4.59 (15.48)	0.30	65.6%
	1 year	25	70.24 (16.74)	11.36 (13.00)	0.87	80.0%
Nutrition	Baseline	97	56.74 (20.29)			
	1 week	36	61.11 (19.56)	1.78 (24.74)	0.07	47.2%
	3 months	35	63.31 (18.75)	3.43 (23.90)	0.14	48.6%
	1 year	26	62.46 (18.93)	4.31 (21.94)	0.20	38.5%